

SAO VOJA4 ADAPTER (with SPI!)

GPIO TO I2C CONVERTER

Voja4 set OUTx HIGH when not in use for tri-state mode.
Voja4 set OUT0 LOW to assert I2C line low.

1st NPN is ON. /OUT is LOW.
1st NPN is OFF. /OUT is HIGH.

2nd NPN is OFF. I2C line is floating HIGH.
2nd NPN is ON. I2C line is asserted LOW.

DUAL SPI, CS1&2

Follow Single SPI, CS1 config.
-AND-
Close JP4.
Open JP2,3,5,6,7.
Connect base pad Q4 to JP30 pad 3.
Remove Q4.
USE: OUT1 HIGH = CS2 LOW

SINGLE SPI, CS1

Close JP10,11,12,13,22,23.
Open JP20,21.
Close JP30 pad 1-2.
USE: OUT1 LOW = CS1 LOW

SAO, I2C, QWIIC

<https://hackaday.io/project/175182-simple-add-ons-sao>
using Sullins SFH11-NBPC-D03-ST-BK female header
<https://www.digikey.com/product-detail/en/sullins-connector-solutions/SFH11-NBPC-D03-ST-BK/S9717-ND/4558818>

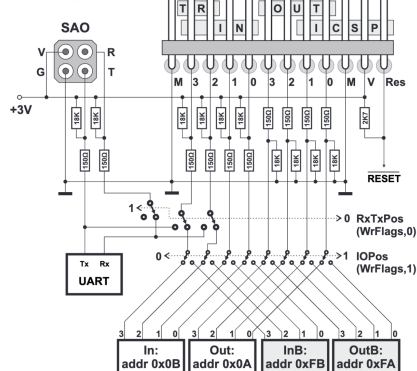
OLED COLOR SPI W/MICRO SD

Compatible: <https://www.adafruit.com/product/1431>
1.5" 128x128, 16-bit color w/MicroSD holder

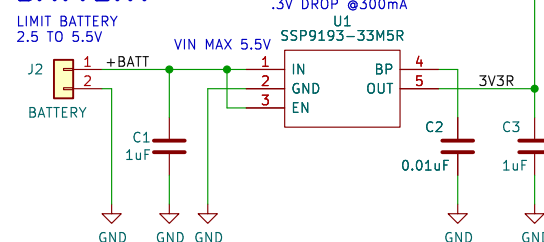
MICRO SD CARD

Compatible with
<https://www.adafruit.com/product/4682>

BADGE



ALTERNATE 3V3 REGULATOR BATTERY



L1
Machine Ideas Logo

All non-polarized capacitors are X7R or X5R ceramic unless otherwise noted.
Designed by: Andy G., Koppany H.

SPI CONFIGURATION

Sheet: /
File: SAO Voja4 Adapter.kicad_sch

Title: SAO Voja4 Adapter

Size: A Date: 2024-10-22

KiCad E.D.A. 8.0.4

Rev: 1.0

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